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ON CERTAIN AVERAGE CHARACTERISTICS OF FINITE CONTINUOUS-TIME MARKOV CHAINS

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We consider a nonhomogeneous continuous-time Markov chain with finite phase space and 1-periodic intensities. The behavior of two important characteristics of the process is studied, namely, we obtain the bounds on the rate of convergence to the limiting mean and to double mean. Some examples of introduced values analysis are considered.

Introduction

The problem of analyzing convergence rate to the limit mode for different classes of Markov chains has been actively studied for the past few decades. In the present paper we consider an inhomogeneous Markov chain $X(t)$ with finite phase space $\Upsilon = \{0, 1, \ldots, N\}$ and 1-periodic intensities. The behavior of two important characteristics of the process are studied, namely, limit average and double average. These characteristics for the inhomogeneous processes of birth and death are introduced and studied in [5, 6]. Rates of convergence to the limit average and double average are estimated, and some examples of introduced values analysis are considered.

A Marcov chain $X(t)$ is called weakly ergodic if

$$\lim_{t\to\infty} \left\| \mathbf{p}^{(1)}(t) - \mathbf{p}^{(2)}(t) \right\| = 0$$

for any initial probability distributions of $\mathbf{p}^{(1)}(0)$, $\mathbf{p}^{(2)}(0)$. Hereafter vector l_1-norm is used, i.e., $\|\mathbf{x}\| = \sum_{i\in\Upsilon} |x_i|$ for $\mathbf{x} = (x_0, \ldots, x_N)^T$.

Definition 1. We call function $\phi(t)$ limit average for the chain $X(t)$, if for any $k \in \Upsilon$

$$\lim_{t\to\infty} |E\{X(t) \,|X(0) = k\} - \phi(t)| = 0.$$

Definition 2. If the limit

$$\mathsf{E} = \lim_{t\to\infty} \frac{1}{t} \int_0^t E\{X(u) \,|X(0) = k\}\, du \tag{1}$$

exists and does not depend on $k \in \Upsilon$, then E is called double average for the chain $X(t)$.

Let $A(t)$ be the intensity matrix of our chain. Hereafter we assume that $A(t)$ is 1-periodic and integrable on $[0, 1]$. Let us introduce "averaged" matrix

$$A_0 = \int_0^1 A(t)\, dt. \tag{2}$$

1. Estimates

Theorem 1. *Let homogeneous Markov chain with intensity matrix A_0 have accessible state. Then Markov chain $X(t)$ with 1-periodic intensity matrix $A(t)$ is weakly ergodic. It has 1-periodic limit mode $\pi(t)$, limit average $\phi(t)$, and double average E. And there exist $\alpha > 0$ and natural m_* such that for all natural n:*

1) *for all $t \geqslant m_* n$ and any $\mathbf{p}(0)$*

$$\|\mathbf{p}(t) - \pi(t)\| \leqslant 2(1-\alpha)^n; \tag{3}$$

2) *for all $t \geqslant m_* n$ and any $k \in \Upsilon$*

$$|E\{X(t) \,|X(0) = k\} - \phi(t)| \leqslant 2N(1-\alpha)^n; \tag{4}$$

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3) *for any* $k \in \Upsilon$

$$\left| \int\limits_{m_* n}^{m_* n+1} E\left\{X(t)|X(0)=k\right\}\, dt - \mathsf{E} \right| \leqslant 2N\left(1-\alpha\right)^n. \tag{5}$$

Proof. Following the scheme proposed in [3, 8], let us first note that if the state i_* of homogeneous chain $X_0(t)$ with intensity matrix A_0 is accessible, then for any $j \neq i_*$ there exists intensity sequence $a_{j,i_1}(t)$, $a_{i_1,i_2}(t), \ldots,$ $a_{i_{R-1},i_*}(t)$ such that for any element of this sequence $\int_0^1 a_{i_m,i_{m+1}}(t)\,dt > 0$, and the sequence length $R \leqslant N$.

Writing out direct Kolmogorov's system for $X(t)$, assuming $s \geqslant 0$, and choosing arbitrarily $X(s) = j$, we obtain

$$\frac{dp_j(t)}{dt} = \sum_k a_{k,j}(t)p_k(t) \geqslant a_{j,j}(t)p_j(t),$$

and $\quad p_j(t) \geqslant \exp\left(\int_s^t a_{j,j}(u)\,du\right) > 0$ for all $t \geqslant s$. Now, following the accessibility sequence, we have

$$\frac{dp_{i_1}(t)}{dt} = \sum_k a_{k,i_1}(t)p_k(t) \geqslant a_{i_1,i_1}(t)p_{i_1}(t) + a_{j,i_1}(t)p_j(t) \geqslant a_{i_1,i_1}(t)p_{i_1}(t) + a_{j,i_1}(t)\exp\left(\int\limits_s^t a_{j,j}(u)du\right),$$

from which

$$p_{i_1}(t) \geqslant \int\limits_s^t a_{j,i_1}(u)p_j(u)\exp\left(\int\limits_u^t a_{i_1,i_1}(\tau)\,d\tau\right)\,du > 0$$

for all $t - s \geqslant 1$, then

$$\frac{dp_{i_2}(t)}{dt} = \sum_k a_{k,i_2}(t)p_k(t) \geqslant a_{i_2,i_2}(t)p_{i_2}(t) + a_{i_1,i_2}(t)p_{i_1}(t),$$

from which

$$p_{i_2}(t) \geqslant \int\limits_s^t a_{i_1,i_2}(u)p_{i_1}(u)\exp\left(\int\limits_u^t a_{i_2,i_2}(\tau)\,d\tau\right)\,du > 0$$

for all $t - s \geqslant 2$ and so on. As a result we have

$$p_{i_*}(t) \geqslant \int\limits_s^t a_{i_{R-1},i_*}(u)p_{i_{R-1}}(u)\exp\left(\int\limits_u^t a_{i_*,i_*}(\tau)\,d\tau\right)\,du > 0$$

for all $t - s \geqslant R$ and strict positivity of all $p_{j,i_*}(t)$ for any j and every $t \geqslant R$. Hence weak ergodicity of $X(t)$ and existence of limit 1-periodic mode of probability distribution for states $\pi(t)$ follows.

Let

$$\alpha_k = \max_j \min_i \mathsf{P}\left(X(k) = j | X(0) = i\right).$$

Note that $\alpha_R > 0$. Let us now choose $m_* = \min\left(k|\alpha_k > 0\right),\ \alpha = \alpha_{m_*}$ Then we have (3). To prove (4) let $\phi(t) = \sum_{j=0}^N j\pi_j(t)$. Then we have

$$\left|E\left\{X(t)\,|X(0)=k\right\} - \phi(t)\right| \leqslant \sum_{j=0}^N j|p_{kj}(t) - \pi_j(t)| \leqslant N\|\mathbf{p}(t) - \pi(t)\| \leqslant$$

$$\leqslant 2N\left(1-\alpha\right)^n. \tag{6}$$

Defining

$$\mathsf{E} = \int\limits_0^1 \phi(t)\,dt, \tag{7}$$

we have that (5) follows from (4).

Remark 1. It is easy to prove that condition of accessible state existence for homogeneous chain with intensity matrix A_0 is necessary for weak ergodicity of $X(t)$.

2. Examples

First let us consider the model described by so-called general birth and death process, which was studied in [2]. This class includes multidimentional processes with several particle types, for which the following transitions have nonzero intensities: incrementing or decrementing by 1 the number of particle of a single type, particle transition from one type to another.

Example 1. Process that describes the model of seasonal epidemy with two types of individuals: healthy (receptive) and sick, with the possibility to increase or decrease the number of each type by one and to transit from one type to another, and maximal number of individuals of each type equal to $N_i,\ i = 1, 2$. We assume that intensities of birth and death for each class are constant and equal to 1, and intensities of infection and recovery are correspondingly $0.5 + 0.4\cos(2\pi t)$ and $0.5 + 0.5\sin(2\pi t)$. Along with $\phi(t)$ and E, which describe average characteristics of the whole population, particular interest lies in corresponding averages $\phi_i(t)$ and $\mathsf{E_i}$ for each class of individuals ($i = 1, 2$). Then for $N_1 = N_2 = 3$ we have $\alpha = 0.09691745$ and to calculate averages with reasonable accuracy (0.000001) one has to take $t > 150$. In this case double averages are $\mathsf{E}_1 = 1.01279192,\ \mathsf{E}_2 = 1.00242102$.

Figure 1 shows graphs of averages $\phi_1(t)$ and $\phi_2(t)$ for $150 \leqslant t \leqslant 151$. When the class sizes N_1 and N_2 increase, α decreases rapidly. Thus for $N_1 = N_2 = 10$ we have $\alpha = 1.09903868 \cdot 10^{-5}$, and for $N_1 = N_2 = 12$ we have $\alpha = 2.98688894 \cdot 10^{-7}$. This points to a very slow convergence to the limit average and double average.

Example 2. To speed up convergence rate in the previous example let us introduce a "drift" to the null state, assuming that transitions from any state to the state $(0; 0)$ are possible with intensity 1, meaning individual withdrawal from population. Then from equation for $p_{00}(t)$ we obtain the estimate $p_{00}(t) \geqslant \frac{1}{3} - \exp(-3t)(p_{00}(0) - \frac{1}{3})$, from which $p_{00}(1) \geqslant \frac{1}{3}(1 - \exp(-3)) \geqslant 0.3167$ and therefore $\alpha > 0.316$ for any $N_1,\ N_2$. In this case to ensure accuracy 0.00001 it is sufficient to take $t > 32$.

Thus for $N_1 = 10,\ N_2 = 10$ we have $\alpha = 0.32242605$ and double averages $\mathsf{E}_1 = 0.69459720,\ \mathsf{E}_2 = 0.70084721$.

Figures 3 and 4 show graphs of averages $\phi_1(t)$ and $\phi_2(t)$ for $40 \leqslant t \leqslant 41$.

Example 3. Consider the queueing system $M(t)/M(t)/S/S$, see, e.g., [1, 4, 7]. The number of claims in this system $X(t)$ is a birth and death process with finite phase space $\Upsilon = \{0, 1, \ldots, S\}$ and intensities

$$\lambda_n(t) = a(t),\ \mu_n(t) = nb(t). \tag{8}$$

First estimates for inhomogeneous case and weak ergodicity criterion for this process were obtained in [3].

Let

$$a(t) = 1 + \sin(2\pi t),\ b(t) = 1 + \cos(2\pi t). \tag{9}$$

Then applying Theorem 1 we find $\alpha = 0.33748171$ for $S = 3$, and to ensure accuracy 0.000001 one has to take $t > 40$. In this case we have double average $\mathsf{E} = 0.86345907$. Figure 5 shows graph of average number of claims ($40 \leqslant t \leqslant 41$).

Taking $S = 9$, we have $\alpha = 0.09225664$, and to ensure accuracy 0.000001, it is necessary to take $t > 150$. In this case we have double average $\mathsf{E} = 1.09205886$. Figure 6 shows graph of average number of claims ($150 \leqslant t \leqslant 151$).

Taking $S = 100$, we have $\alpha \approx 0$, and Theorem 1 does not give the opportunity to find average characteristics of the system.

Much better estimates for such models may be obtained using results of [5, 6]. From Theorems 1–3 of [6] we have the following estimates (choosing auxiliary sequence $d_i = 1,\ i \geqslant 1$):

$$|E\{X(t)\,|X(0) = 0\} - \phi(t)| \leqslant e^{4-t} \tag{10}$$

and

$$\left| \int_n^{n+1} E\{X(t)|X(0) = 0\}\, dt - \mathsf{E} \right| \leqslant e^{4-t} \tag{11}$$

for any $t \geqslant 0$.

It means that to ensure accuracy 0.000001 it is sufficient to take $t \geqslant 20$ for any S. Thus for $S = 100$ we can use this estimate and to obtain double average $\mathsf{E} = 1.09206972$. Graph of average is shown in Fig. 7.

3. Appendix

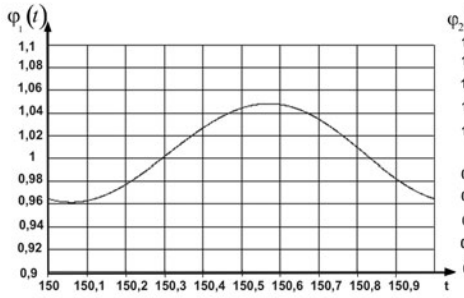


Fig.1

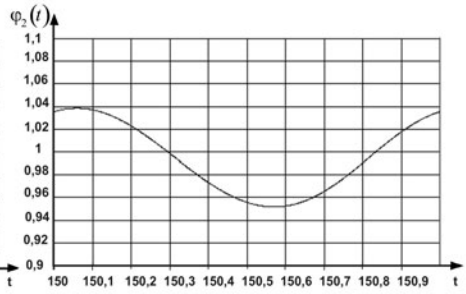


Fig.2

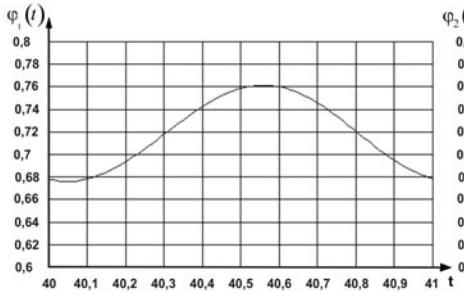


Fig.3

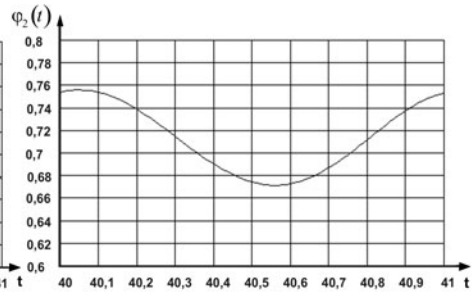


Fig.4

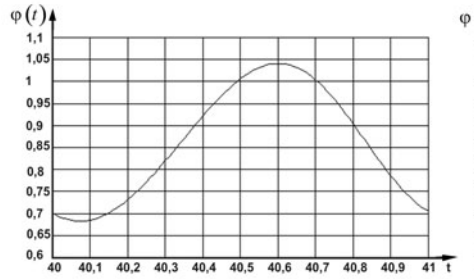


Fig.5

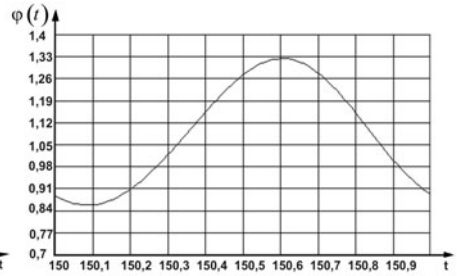


Fig.6

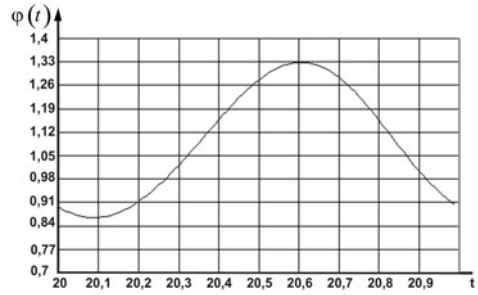


Fig.7

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